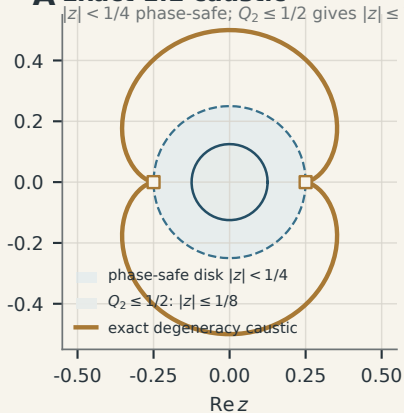


PHASE-ROBUST MORSE SHAPE · EXACT CONTRACT

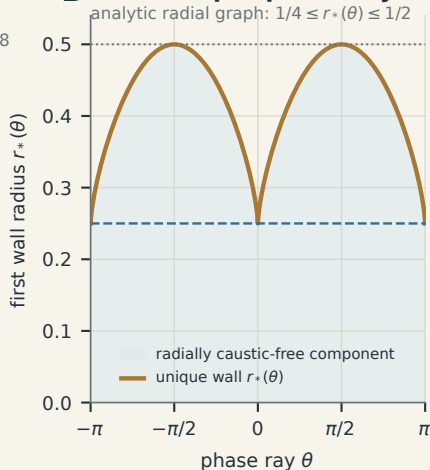
Fixed M and $Q_2 \leq 1/2$ give arbitrary-phase critical-point control; the 1:2 caustic is exact.



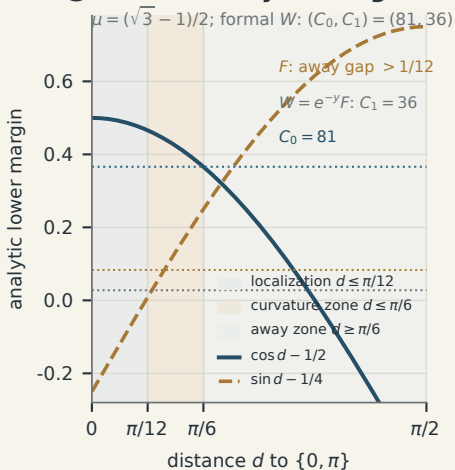
A Exact 1:2 caustic



B One wall per phase ray



C Fixed-M analytic margins



All curves are samples of exact formulas for display; they cannot replace the continuous proof.
Formal Coble shear: $W = e^{-y}F$ on $0 \leq y \leq 1$, with $(r, C_0, C_1) = (\pi/12, 81, 36)$.